

H0: Men and women prefer cats and dogs equally. The preference is independent of gender. H1: The preference for cats and dogs is dependent on gender.

	Cat	Dog
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Women	231	242

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With totals

	Cats	Dogs	Total
Men	207	282	489
Women	231	242	473
Total	438	524	962

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Expected values

	Cats	Dogs	
Men	$\frac{489 \times 538}{962}$	$\frac{489 \times 524}{962}$	489
Women	$\frac{473 \times 438}{962}$	$\frac{473 \times 524}{962}$	473
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Expected values - computed

	Cats	Dogs	
Men	222.64	266.36	489
Women	215.36	257.64	473
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Expected values - computed

	Cats	Dogs	
Men	222.64	266.36	489
Women	215.36	257.64	473
	438	524	962

	Cats	Dogs	
Men	$\frac{(207-222.64)^2}{222.64}$	$\frac{(282-266.36)^2}{266.36}$	489
Women	$\frac{(231-215.36)^2}{215.36}$	$\frac{(242-257.64)^2}{257.64}$	473
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	Cats	Dogs	
Men	$\frac{(207-226.4)^2}{226.4}$	$\frac{(282-266.36)^2}{266.36}$	489
Women	$\frac{(231-215.36)^2}{215.36}$	$\frac{(242-257.64)^2}{257.64}$	473
	438	524	962

	Cat	Dog
Men	1.099	0.918
Women	1.136	0.949

Add them up and that is the X^2 (Chi-squared or Chi-square) score:

$$X^2 = 1.099 + 0.918 + 1.136 + 0.949$$

$$X^2 = 4.102$$

Now we take it to the table or to the chi-square distribution function in R, but first we have to know a new number:

Degrees of Freedom

Degrees of freedom is a lot like what it sounds like. It's that maximum number of logically independent variables. It's essentially the "wiggle room" - how much do the variables allow for a good measurement?

In Chi Square, the degrees of freedom is:

$$df = (rows - 1) \times (columns - 1)$$

In this case, $(2-1) \times (2 - 1) = 1$

Second Example

UC Berkley Graduate Admissions, 1973, 6 largest programs

H0: Admissions are independent of gender.

H1: Admissions are dependent on gender.

(The "totals" are also referred to as *Marginal Frequency*.)

Table 1: Values

	Male	Female	Marginal Freq.
Admit	1198	597	1755
Reject	1493	1278	2771
Marginal Freq.	2691	1835	4526

Table 2: Expected values

	Male	Female	Marginal Freq.
Admit			1755
Reject			2771
Marginal Freq.	2691	1835	4526

Table 3: *Observed – Expected*

	Male	Female	Marginal Freq.
Admit			1755
Reject			2771
Marginal Freq.	2691	1835	4526

Table 4: $\frac{(Observed - Expected)^2}{Expected}$

	Male	Female	Marginal Freq.
Admit			1755
Reject			2771
Marginal Freq.	2691	1835	4526

Add it up:

Degrees of freedom: $(2 \times 1) - (2 \times 1) = 1$

p-value: